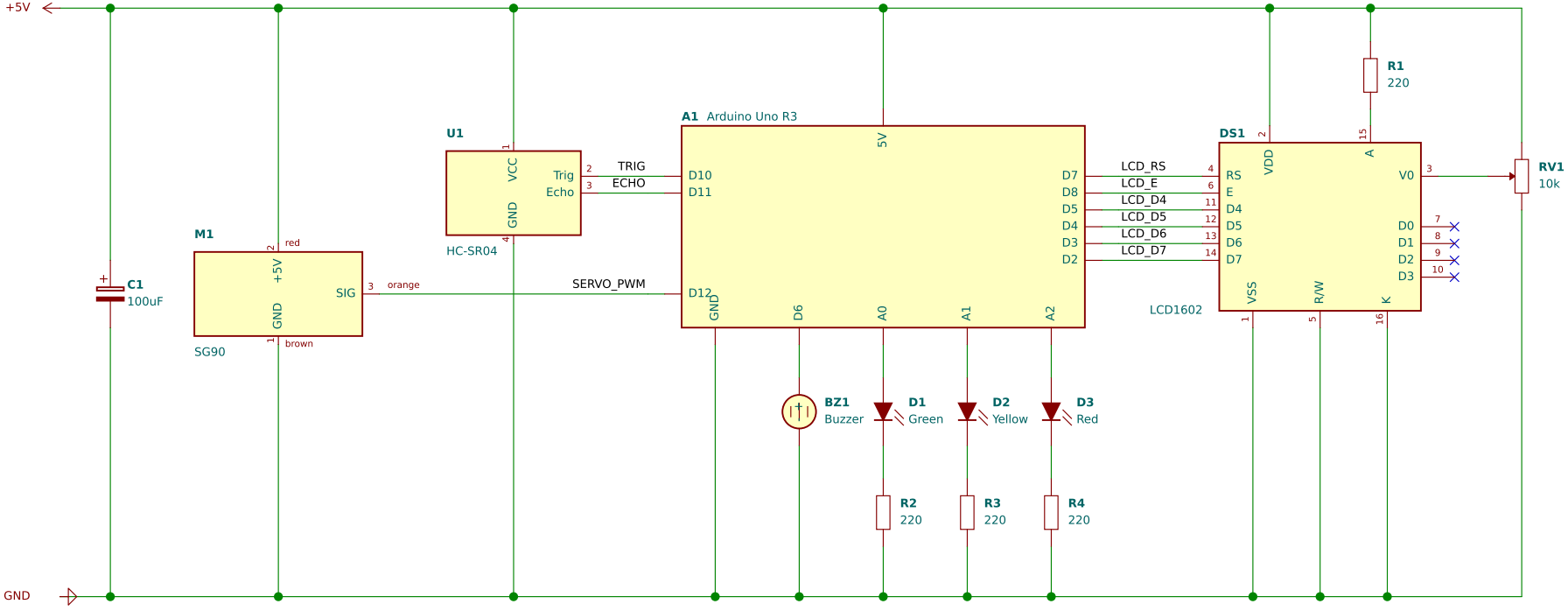
Arduino Ultrasonic Radar - full circuit (Arduino Uno R3)

Top wire = breadboard + rail (Uno 5V). Bottom wire = breadboard - rail (Uno GND).

A dot means wires are joined. Wires that cross without a dot are NOT connected.



Power: 9 V battery into the Uno barrel jack (or USB). The rails always carry the Uno's regulated 5 V, never the raw 9 V. C1 sits close to the servo wires to absorb current spikes.

HC-SR04 is mounted on the servo horn. LEDs: green > 50 cm, yellow 20-50 cm, red < 20 cm.

RV1 sets LCD contrast (V0 about 1.2 V). LCD D0-D3 unused (4-bit mode). D0/D1 on the Uno left free for USB serial.